Stats can be fun, I swear

Picking the question

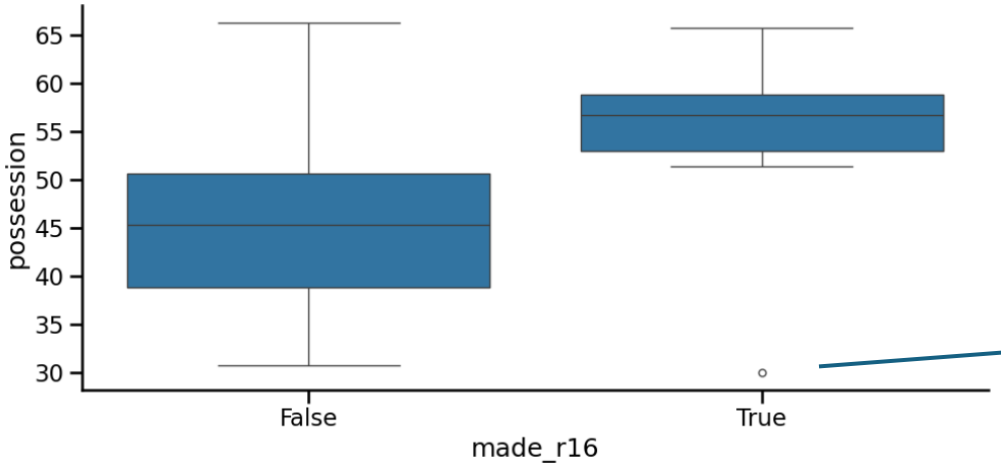
- Before we start dealing with numbers, we need to know what we're actually trying to come to a conclusion on
- Our question needs to be something:
  - Relevant and able to be answered by the dataset
  - Worth finding out

Let's look at this world cup team dataset. One question we could ask is:

What factors influence a country's success?
How can we predict if a country will be successful at the World Cup?

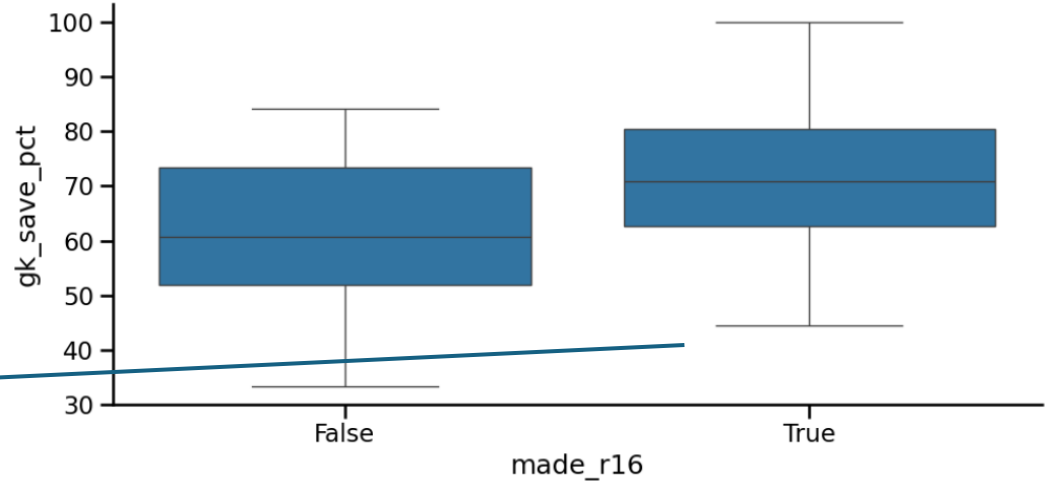
Exploratory data analysis

- This just means making lots of graphs to try and figure out what factors are important to a team's success, and where there are (or aren't) correlations



Box plots for 1 discrete variable

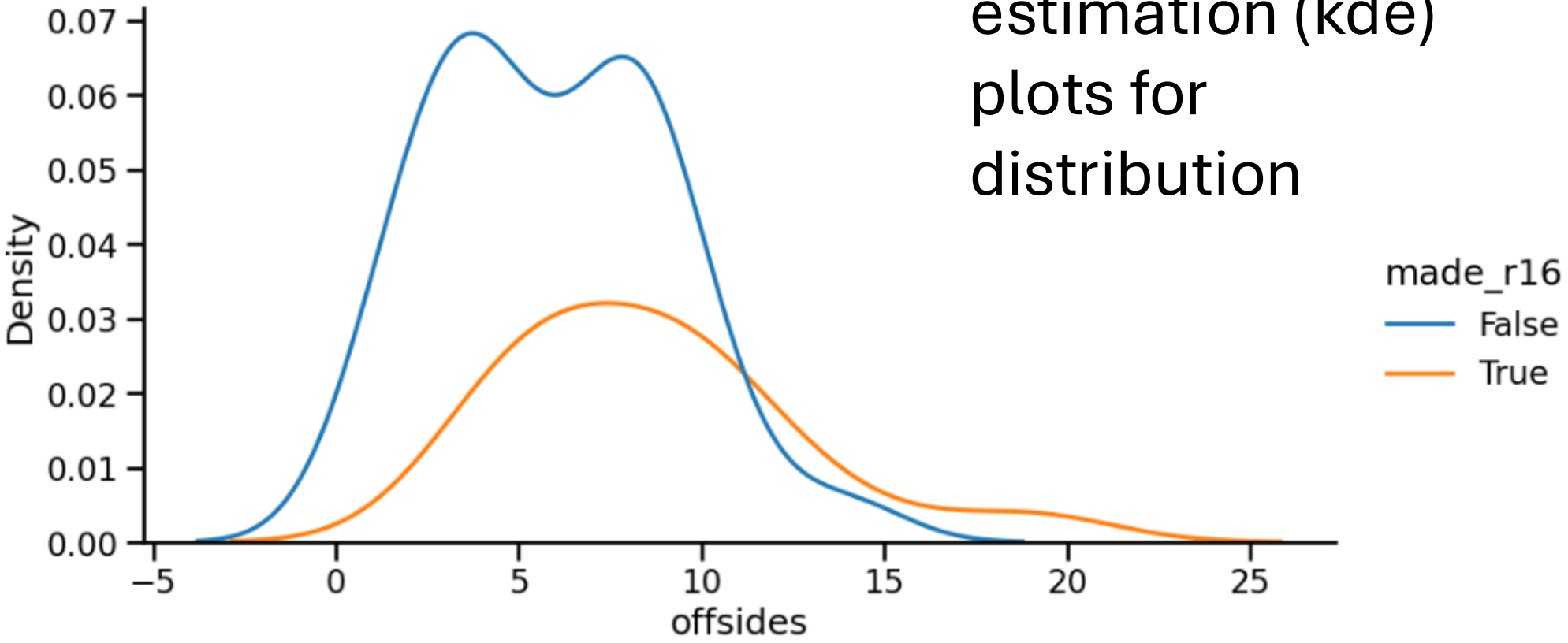
Paraguay made R16 with just 30% possession!



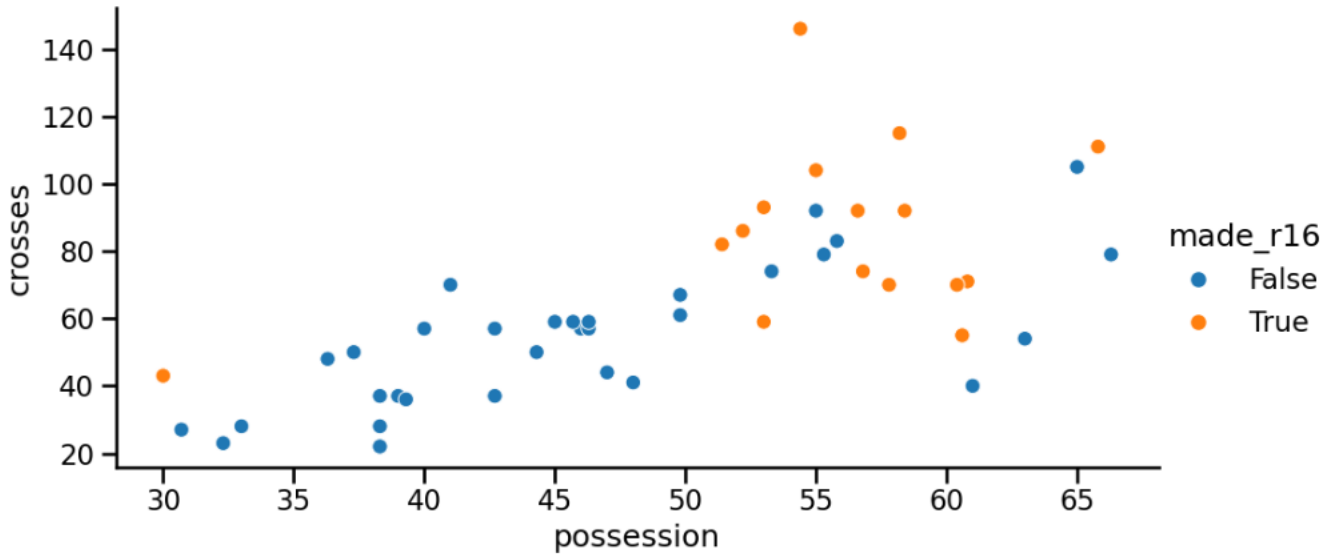
This boundary value is, interestingly, Argentina. shows you how heavy they are on attack and messi (which the rest of their stats show)

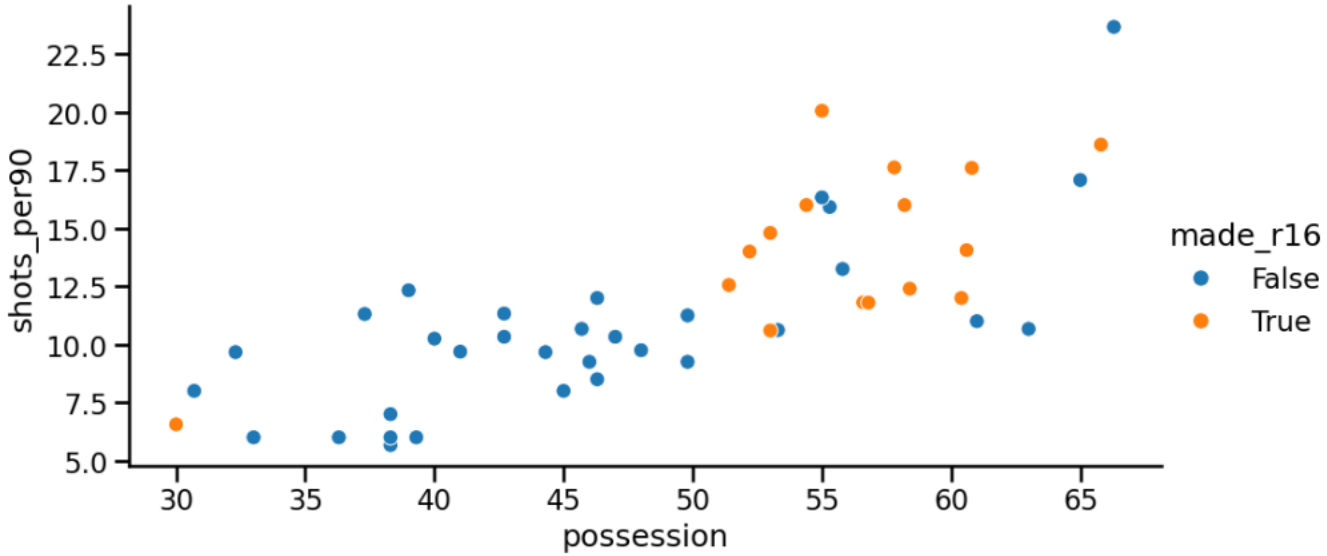
Exploratory data analysis

Kernel density estimation (kde) plots for distribution



Exploratory data analysis

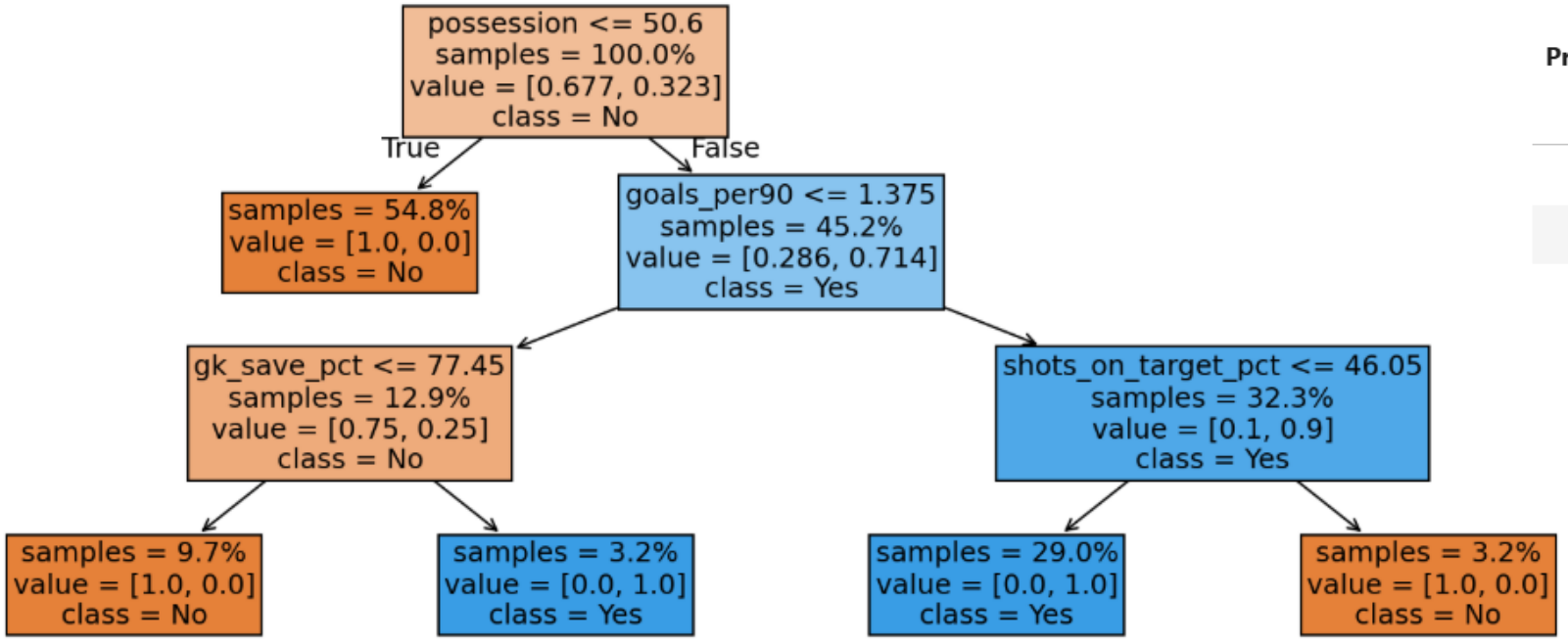




Scatter plots for 2 variables; possession seems to correlate w/ lots of variables, so we want to include it in any models

Binary classification models

- Binary i.e. two outcomes
- If we take success to mean, say, making the round of 16, then this is a great way to classify it; you either make R16 or not.



Predicted	False	True	All
Actual			
False	9	2	11
True	2	4	6
All	11	6	17

13/17 correct predictions; not bad..

Linear regression models

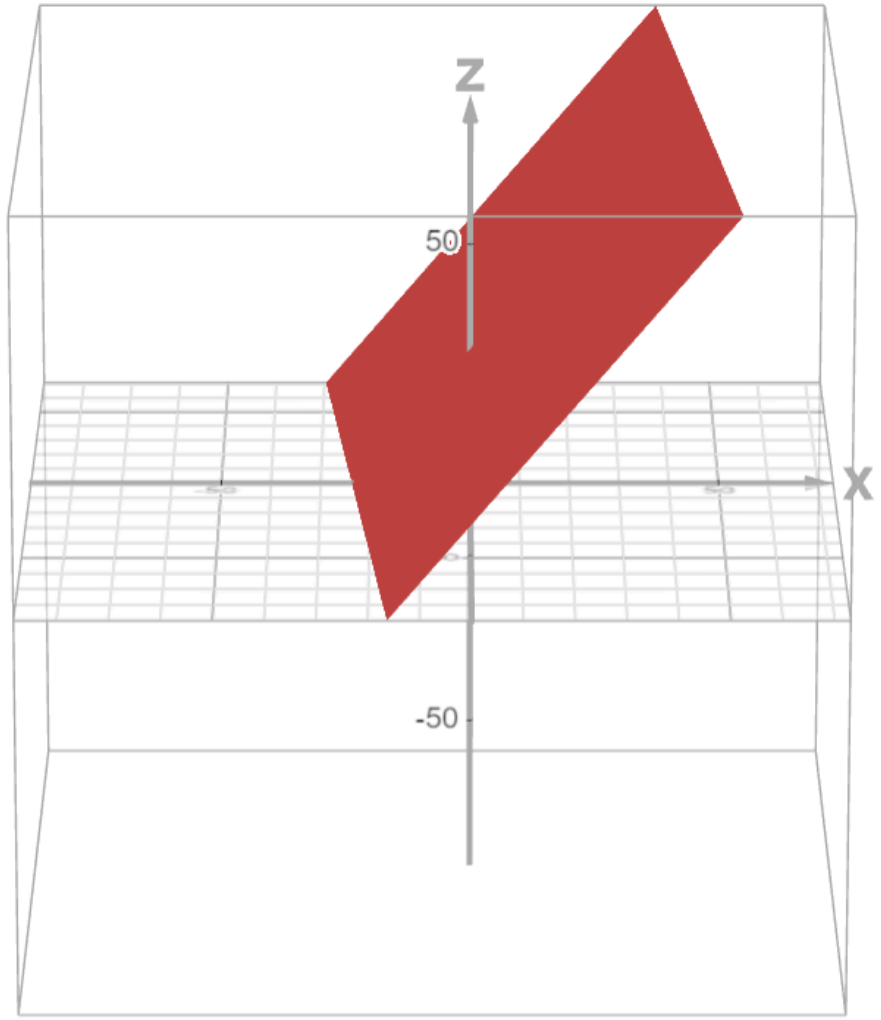
- Hello A-level maths… 
- Useful when we want to predict numerical outcomes. Let say we wanted to predict possession % based on a number of factors
- At a-level, we can do that with one value. In real life, we use several variables to determine one output.
- A similar method, known as MRP, is used in advanced election prediction models

Linear regression models

When we use shots per 90 mins and crosses as our input variables:

```
Coefficients:
 [1.186 0.098]
Intercept:
 28.215
RMSE:  42.619
R²:  69.66 %
```

This actually isn't a very good model.. You want RMSE to be as low as possible.



Analysing data

- You obviously can’t predict everything about how successful a team will be from numbers. Things like:
  - Player experience in tier 1 club football
  - Manager/coaching staff
  - Injuries

All have impacts we can’t quantify.

- What’s more, the dataset isn’t perfect
  - It ignores historic performance, standout performances, unexpected losses
  - Is a really quite small sample; again we ought to mix in the past 2 or 3 world cups too to get something more representative
  - I picked out a select few factors I thought might have worked as else this table had 136 columns. So I might’ve missed a few – clean sheet % immediately comes to mind!
  - Source of the data? Is it reliable?

In summary,

the A-level maths spec is pretty rubbish and doesn't really understand what stats is for.

Stats can be fun and is super useful in lots of scenarios

Yay stats ☺

